MRPL CDU-1 | HEAT EXCHANGER E-204 INSPECTION REPORT

Equipment Integrity & Inspection Report

Author: SOVA Sovereign Intelligence | **Date:** 2026-09-26 | **Classification:** Confidential - Air-Gapped Internal

EXECUTIVE SUMMARY

This report summarizes the inspection results for Heat Exchanger E-204 at MRPL Refining Complex - CDU-1. The inspection revealed a warning at Nozzle N3 and a normal condition at other components. The report provides recommendations for immediate operating pressure review, inspection interval reduction, mitigation, online corrosion monitoring, and engineering review sign-off.

1.0 General Equipment Specification

The equipment is a TEMA Class R - Shell and Tube Heat Exchanger (AES Type) with a design code of ASME Section VIII Division 1 (2019 Edition) / TEMA Class R.

The design pressure is 15.5 bar (g) on the shell side and 18.0 bar (g) on the tube side, with a design temperature of 185 deg C on the shell side and 110 deg C on the tube side.

- Equipment Tag: E-204
- Equipment Type: TEMA Class R - Shell and Tube Heat Exchanger (AES Type)

1. The equipment is subject to API 510 (Pressure Vessel Inspection) & API 579-1 / ASME FFS-1 inspection standards.

Table: UTM Results

Component Location	Nominal (mm)	Min Req t_min (mm)	Measured (mm)	Status
Shell Course 1 (Inlet Nozzle N3)	10.0	7.8	8.2	WARNING
Shell Course 2 (Center Ring)	10.0	7.8	9.1	NORMAL
Shell Course 3 (Outlet)	10.0	7.8	9.4	NORMAL
Channel Head (Shell Side)	12.0	9.2	11.4	NORMAL
Floating Head Cover	14.0	10.5	13.2	NORMAL
Stationary Tubesheet	25.0	20.0	24.6	NORMAL
Nozzle N1 (Crude Inlet 10-inch)	9.27	6.5	8.6	NORMAL
Nozzle N3 (Vapor Return 8-inch)	8.18	5.8	6.4	WARNING

2.0 Ultrasonic Thickness Measurement (UTM) Results

The UTM results showed a warning at Nozzle N3 with a measured thickness of 8.2 mm, which is below the minimum required thickness of 7.8 mm.

- Component Location
- Nominal (mm)
- Min Req t_min (mm)
- Measured (mm)
- Status

1. Shell Course 1 (Inlet Nozzle N3)
2. Shell Course 2 (Center Ring)
3. Shell Course 3 (Outlet)
4. Channel Head (Shell Side)
5. Floating Head Cover
6. Stationary Tubesheet
7. Nozzle N1 (Crude Inlet 10-inch)
8. Nozzle N3 (Vapor Return 8-inch)

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